

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Third Semester B.Tech (E&E) (CBCS) Degree Examinations

Summer Semester July 2024

EE2101-1 – ANALOG SIGNAL PROCESSING

(EE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II b** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- What is the value of CMRR of an ideal opamp?
 - Zero
 - Infinite
 - Unity
 - 0.5
- Opamp is a
 - Voltage controlled voltage source
 - Voltage controlled current source
 - Current controlled voltage source
 - Current controlled current source
- An non inverting amplifier with $A_{CL} = 50$ is fed with 1V input signal. If the circuit uses 741 IC with 15 dual supply, the output of the amplifier is
 - 50
 - 15
 - 15
 - 50
- When a large sine wave drives a Schmitt trigger, the output is a
 - Triangular wave
 - Rectangular wave
 - Rectified sine wave
 - Series of ramps
- The condition which ensure MOSFET operating in saturation region is
 - $V_{DS} < V_{OV}$
 - $V_{GD} < V_{th}$
 - $V_{GS} < V_{th}$
 - $V_{GS} < V_{OV}$
- N channel MOSFET is an example for
 - Majority carrier with electron as carrier
 - Minority carrier with electron as carrier
 - Majority carrier with hole as carrier
 - Minority carrier with hole as carrier
- A given MOSFET is operating in saturation with drain current I_D . If V_{DS} value is doubled the new value of drain current is
 - $2I_D$
 - $0.5I_D$
 - $4I_D$
 - I_D
- MOSFET operating in saturation is related to the internal structure by
 - Inversion layer
 - Pinch off
 - Uniform channel
 - Silicon oxide layer
- MOSFET under saturation acts as
 - Voltage controlled voltage source
 - Voltage controlled current source
 - Current controlled voltage source
 - Current controlled current source
- For a given MOSFET the operating point Q is decided by
 - V_{GS}
 - I_D
 - V_{DS}
 - V_{th}
- The biasing technique best suited for IC application is
 - Fixed V_{GS} biasing
 - Biasing with fixed V_G and R_s
 - Constant current source biasing
 - Biasing with gate to drain resistor
- Voltage gain of the CS amplifier decided by
 - g_m
 - V_{DS}
 - I_D
 - V_{th}

13. Ideal gain of common drain amplifier
 - A) Infinite
 - B) Unity
 - C) Zero
 - D) 10
14. CS amplifier has
 - A) High input impedance low output impedance
 - B) Low input impedance low output impedance
 - C) Low input impedance high output impedance
 - D) High input impedance high output impedance
15. The MOSFET amplifier having less input impedance is
 - A) CS amplifier
 - B) CG amplifier
 - C) CD amplifier
 - D) CS amplifier with source resistance
16. The high frequency response of CS amplifier depends upon
 - A) Internal capacitor
 - B) Coupling capacitor
 - C) Bypass capacitor
 - D) Stray capacitor
17. RC phase shift oscillator is an example for
 - A) Negative feedback system
 - B) Unity feedback system
 - C) Positive feedback system
 - D) Open loop system
18. CS amplifier is an example for
 - A) Shunt-Series feedback
 - B) Series-Series feedback
 - C) Series-Shunt feedback
 - D) Shunt-Shunt feedback
19. Ideal common mode gain of differential amplifier is
 - A) infinite
 - B) zero
 - C) unity
 - D) 100dB
20. The gain bandwidth product of feedback system is always
 - A) Unity
 - B) Zero
 - C) Constant
 - D) None of these

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) Explain working principle of high Z_{in} capacitor coupled non inverting amplifier and describe the design steps followed. | 8 | L*2 | 1 | 3 |
| b) With neat cross sectional view describe the internal structure of MOSFET for various operating regions. | 8 | L2 | 2 | 3 |
| 2. a) With neat circuit diagram explain working of inverting Schmitt trigger circuit. | 6 | L2 | 1 | 3 |
| b) With appropriate equations for drain current, describe the operating regions of MOSFET. | 5 | L2 | 2 | 3 |
| c) Consider an NMOS transistor fabricated in a 0.8- μ m process with $L = 0.8\mu\text{m}$, $W = 16\mu\text{m}$, $t_{ox} = 15\text{nm}$, $\mu_n = 500\text{ cm}^2/\text{V s}$. Find V_{OV} required to operate MOSFET in saturation with $I_D = 0.1\text{mA}$. | 5 | L3 | 2 | 3 |
| 3. a) Design a 741 based capacitor coupled inverting amplifier with unipolar supply voltage. The specifications of the circuit are $V_{CC} = +15\text{V}$, $V_o = 2.5\text{V}$, $A_{CL} = 50$, $f_L = 150\text{Hz}$, $R_L = 1\text{k}\Omega$. | 8 | L3 | 1 | 3 |
| b) With the help of Voltage transfer characteristics, graphically obtain the expression for various operating region. | 8 | L2 | 2 | 3 |

Unit – II

- | | | | | |
|--|---|----|---|---|
| 4. a) Elaborate DC biasing of MOSFET using fixed V_{GS} method and explain its drawback. | 5 | L1 | 3 | 3 |
| b) Describe biasing of MOSFET by fixed V_G and source resistance method. | 5 | L2 | 3 | 3 |
| c) A CS amplifier utilizes a MOSFET biased at $I_D = 0.25\text{ mA}$ with $V_{OV} = 0.25\text{ V}$ and $R_D = 20\text{ k}\Omega$. The device has $V_A = 50\text{ V}$. The amplifier is fed with a source having $R_{sig} = 100\text{ k}\Omega$, and a $20\text{ k}\Omega$ load is connected to the output. Find R_{in} , A_{vo} , R_o , A_v and G_v . | 6 | L3 | 4 | 2 |

5. a) For MOSFET circuit shown in figure 5a, the V_{DD} is 5V, threshold voltage V_{th} is 0.8V, k_n is 0.5mA/V^2 and $V_D = 1.6\text{V}$. Find value of R.

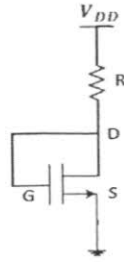


Figure 5a

- b) Describe working of constant current source biasing of MOSFET.
 c) Describe amplifier characteristics of Common Drain amplifier.
6. a) Deduce the T equivalent circuit of small signal model of MOSFET.
 b) Describe gate capacitance and junction capacitances of MOSFET.
 c) With necessary equivalent circuit describe unity gain frequency of Common Source amplifier.

4	L3	3	3
4	L2	3	3
8	L2	4	2
4	L1	3	3
4	L2	4	2
6	L2	4	2

Unit – III

7. a) With neat diagram explain RC phase shift oscillator.
 b) With necessary equivalent circuit describe high frequency response of CS amplifier.
8. a) With necessary equivalent circuit elaborate common mode gain and CMRR.
 b) Elaborate properties of a Negative feedback system.

8	L1	5	3
8	L2	5	3
8	L2	5	3
8	L1	5	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
